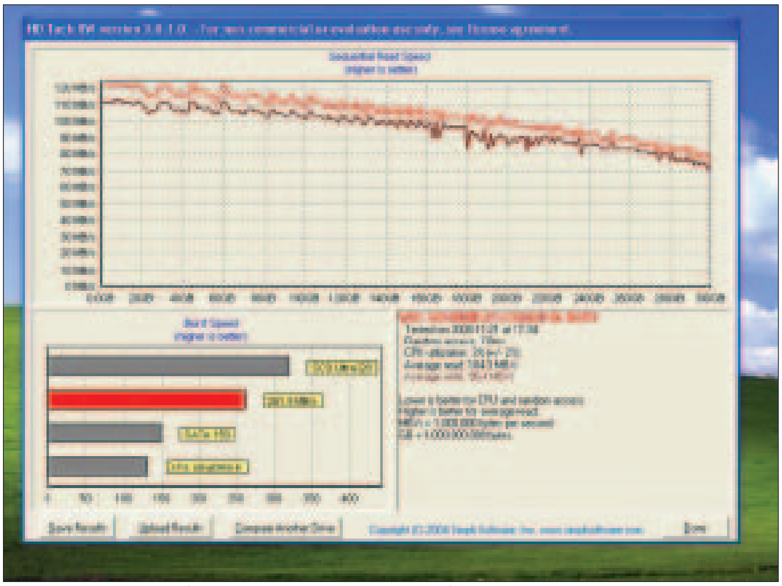

HD Tach — WD Velociraptor; note the excellent write speeds; read speeds are lower than SSD

interface imposes its write speed is pretty ordinary. In comparison the Velociraptor manages a modest read speed 104.3 MBps; but its write speed shines at 96.4 MBps; which is much more than the SSD's write speed of 77.8 MBps. However, since most of your hard drive operations are read and not write operations; take this with a pinch of salt.

We see the SSD ahead of the Velociraptor in all tests; that's a comprehensive victory in the real world scenario. Of course the write tests are a close thing; and while HD Tach lists the Velociraptor as having a higher write speed; the real world tests take into consideration into access time and spin up/down of the spindle as well as mechanical latencies.

A Comparison Of Both Drives In Real-world Tests

Benchmarks (Less is Better)	Intel 80 GB SSD	Western Digital Velociraptor 300 GB
4 GB External write (Sequential)	57.9 sec	77.4 sec
4 GB Internal read / write (Sequential)	73.9 sec	79.5 sec
4 GB External write (Random)	83.3 sec	89.8 sec
4 GB Internal read / write (Random)	78.9 sec	72.3 sec
WinRAR 3.8 Create Archive (4 GB)	77.7 sec	84.3 sec
Adobe PhotoShop CS3 load time (200 MB file)	7.13 sec	12.5 sec
Far Cry (Mission 1) loadup time (sec)	17.8 sec	28.1 sec
STALKER (level Garbage) load time (sec)	21.4 sec	36.7 sec

How We Tested

The following test components were used:

Components	Model Number
Processor	Intel Core 2 Quad Extreme QX 9650
Motherboard	ASUS Rampage Formula (X48)
RAM	Corsair Dominator 1x4 GB, DDR2 1066 MHz
HDD	WD Raptor 80 GB (Primary)
Graphics Card	NVIDIA GeForce 8800 GTX
OS	Windows XP SP2

We installed Windows on the primary drive which was an 80 gigabyte WD Raptor. The test drives were formatted using NTFS file system and the cluster size was kept constant at 4096 bytes.

We used HD Tach 3.0 for theoretical testing; the hard drives to be tested weren't formatted so that we could run HD Tach's write tests which only work if the hard drive being tested is unformatted. In our external tests we copied data from another hard drive to the test drives and the internal test consisted of an inter partition copy.

Another write test involved using WinRAR 3.8 We created a 4 GB archive with "storage" as a setting. This ensures minimal CPU usage since there is no compression. Using Adobe PhotoShop CS3 we opened a 200-MB test file and recorded the time taken for the same. This checks the sequential read speed of the drive, very important for anyone working with large files like 3D renders and image files.

Our last test involves checking the read speed of the drive while loading games. This is the best example of a random read; because in order to play a game multiple files of different sizes and types have to be loaded. *Far Cry* and *S.T.A.L.K.E.R.* are two popular games with extremely long load times; the latter in particular can take as much as a minute to load on a slow PC.

Once all these factors are considered SSD is clearly faster. The great news is that this SSD is flash based; RAM based SSDs will be considerably faster. As faster flash memory based SSDs arrive; you can expect them to push the SATA 3 standard to market. This standard offers 600 MBps transfers.

So SSD is unanimously faster; but a lot costlier as well. An 80 GB SSD costs in the region of Rs 30,000; while a 250 GB SSD will cost close to a lakh, if not more. However, prices could tumble very fast if these products suddenly catch the focus of manufacturers; right now they're pretty much on the fringes.

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